

(a) In the database context, what do we mean by redundant data?[1 mark]

(b) Why might it be a good idea to have redundant data in a database? [1 mark]

(c) Why might it be a bad idea to have redundant data in a database?[1 mark]

a) Data redundancy in database is when there is multiple records of the same data. Basically when data is stored in two or more separate places.

b) It might be good, because then if the primary one become corrupt or unrecoverable then having a back-up/second one can be used

c) It is a bad idea because when you modify data (update/insert/delete), then you need to do it on multiple places on the database. This makes it that the data becomes inconsistent across the database.

We consider the following relation:

Articles(ID,title,journal,issue,year,startpage,endpage,TR-ID)

It contains information on articles published in scientific journals. Each article has a unique ID, a title, and information on where to find it (name of journal, what issue, and on which pages).

Also, if results of an article previously appeared in a "technical report" (TR), the ID of this technical report can be specified.

We have the following information on the attributes:

- For each journal, an issue with a given number is published in a single year.
- The endpage of an article is never smaller than the startpage.
- There is never (part of) more than one article on a single page.

The following is an instance of the relation:

ID	title				journal	issue	year	startpage	endpage	TR-ID
42	Cuckoo	Hashing			JAlg	51	2004	121	133	87
33	DeterministicDictionaries				JAlg	41	2001	69	85	62
33	DeterministicDictionaries				JAlg	41	2001	69	85	56
39	Dictionaries	in	less	space	SICOMP	31	2001	111	133	47
57	P	vs	NP	resolved	JACM	51	2008	1	3	99
77	What	Godel	missed		SICOMP	51	2008	1	5	98
78	What	Godel	missed		Nature	2222	2008	22	22	98

Based on the above, indicate for each of the following sets of attributes whether it is a key for Articles or not. (You need to check the checkbox if it is a key - other case leave it empty!)

- ☐ a. {ID,TR-ID}
- ☒ b. {ID}
- ☒ c. {startpage,journal,issue}
- ☒ d. {title}
- ☐ e. {title,year}
- ☐ f. {ID,title,TR-ID}

Discuss the characteristics of relations that make them different from ordinary tables and files. [2 marks]

Table File A file/table contains a physically stored information of records (on a disk or memory) that are related to each other. The records are physically stored in a file and therefore the ordering affects the performance of the file. The values inside the record are physically stored in the file. The first value in a record is considered as the value of the first attribute and so on. The tables can be created without the not null constraint and saved as a file. Tables and files have some particular order, but relations don't.

Relation: The tuples are not considered to be ordered, even though they appear to be in the tabular form. Ordering of rows does not affect the performance of the operations that are performed on a relation

Karl Andersson has a boat rental. As summer season approaches, he thinks of the mess of paper that he has to deal with every year. It would be so much more convenient to have everything organized digitally. His friend comes up with a first draft of an ER model shown in Figure 1.

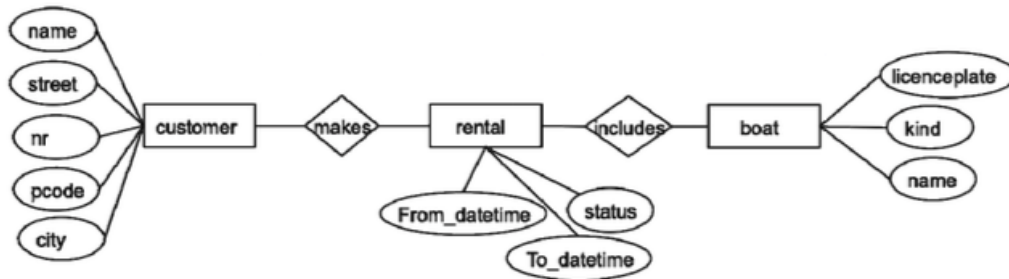


Figure 1. ER model

Every boat has a license plate that identifies the boat, a name, and a type (sailing-, motor boat etc.). Furthermore, Karl stores the customer's name and address (pcode is the postal code). Every rental has a number. A boat is rented from a certain date and time to a certain date and time. The status stores the information whether the boat has been returned or not.

- Mark the primary key in the ER model (adjust the model if necessary) and give a short explanation, why the primary keys are possible to use. (2 pts)
- Add cardinality ratios and briefly explain your decision. (3 pts)
- Translate the ER model to the relational model. For the relationship types: Do you need an extra table? Why or why not? (5 pts)

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Based on the above, perform normalization into BCNF, and state the resulting relations. (5 marks)

journal issue→year

ID→title journal issue startpage endpage

we get the following relations

Articles1(journal,issue,year)

Articles2(ID,title,journal,issue,startpage endpage)

Articles3(ID,TR-ID)

The following relation schema can be used to register information on the repayments on micro loans.

Repayment(borrower_id,name,address,loanamount,requestdate,repayment_date,request_amount)

A borrower is identified with an unique borrower_id, and has only one address. Borrowers can have multiple simultaneous loans, but they always have different request dates. The borrower can make multiple repayments on the same day, but not more than one repayment per loan per day.

a) State a key (candidate key) for Repayment. [1 mark]

b) Make the normalization to BCNF. State for every step in the normalization, which functional dependency that causes it.(5 marks)

a) {borrower_id,requestdate,repayment_date}

b)

borrower_id→name address

borrower_id requestdate→loanamount

bcnf:

Repayment1(borrower_id,name,address)

Repayment2(borrower_id,requestdate,loanamount)

Repayment3(borrower_id,requestdate,repayment_date,repayment_amount)

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a) {borrower_id , requestdate , repayment_date}

b)

borrower_id -> name address

borrower_id requestdate -> loanamount

bcnf:

Repayment1(borrower_id,name,address)

Repayment2(borrower_id , requestdate , loanamount)

Repayment3(borrower_id , requestdate , repayment_date,repayment_amount)

Define properly what is a database! (1 mark)

Database is a collection of related data or operational data extracted from any firm/organization/company /business. A collection of data is called database.

When is the concept of a weak entity used in data modeling? Define the terms owner entity type, weak entity type, identifying relationship type, and partial key. [4 marks]

The concept of a weak entity is used in the conceptual phase of a data modeling. While modeling, the entity types who do not have key attributes of their own.

An example would be, consider the entity types DEPEND and EMPL

DEP can only be an EMPL of the company.

The DEP attributes can be same for relatives of two employees so, there can be no unique way of distinguishing between two records such entity types are called weak entity types.

Owner entity type

The entities belonging to a weak entity type are identified by being linked to specific entities belonging to another entity type and one of their attribute values.

5.5.

Why do we designate one of the candidate keys of a relation to be the primary key?

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78What	Godel	missed		Nature	2222	2008	22	22	98

Based on the above, indicate for each of the following potential functional dependencies, whether it is indeed an FD or not.

(You need to check the checkbox if it is a FD - other case leave it empty!)

- ☐ a. title $\rightarrow$ ID
- ☐ b. startpage $\rightarrow$ endpage
- ☐ c. ID $\rightarrow$ startpage endpage journal issue
- ☐ d. TR-ID $\rightarrow$ ID
- ☐ e. ID $\rightarrow$ title
- ☐ f. journal issue $\rightarrow$ year